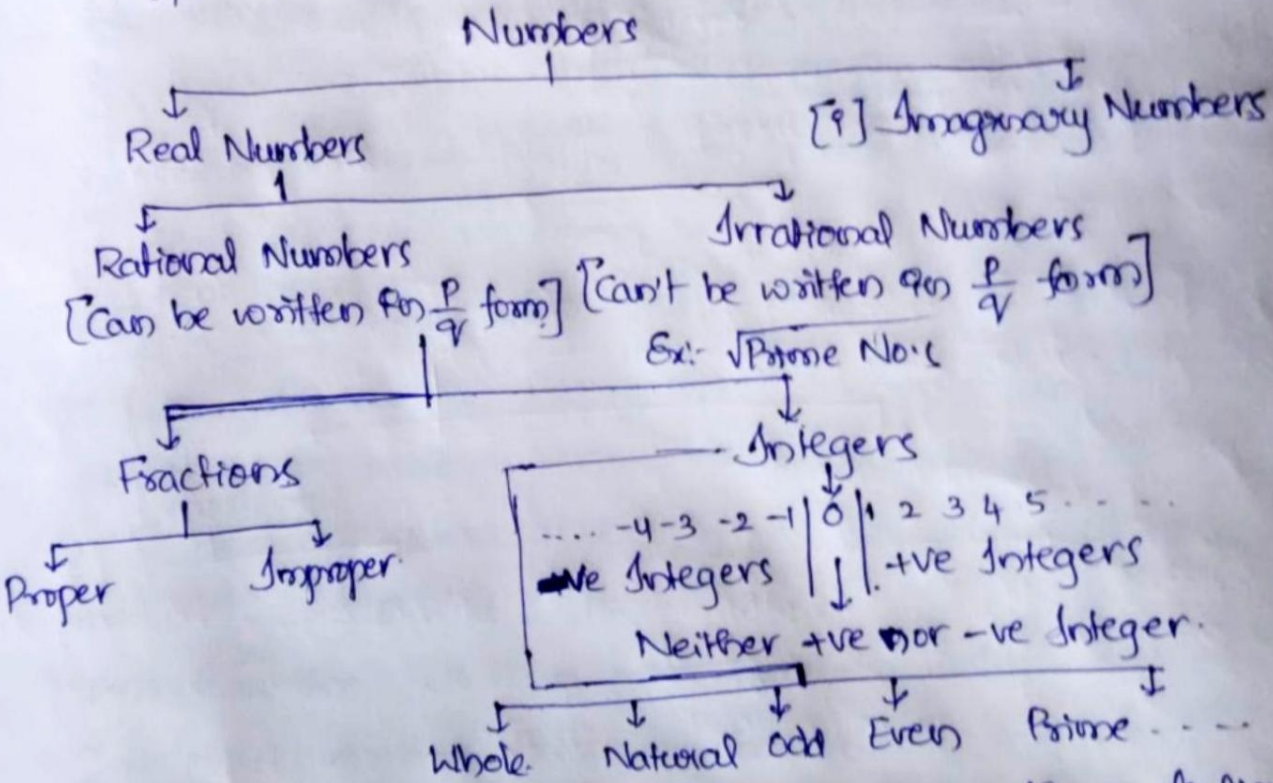
Number Systems (or) Real Number Systems

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↳ Different kind of Applications on numbers to use diff kind of operations as "Number Systems".

↳ Types of Numbers



↳ In general as Number Systems we are seeing the applications on real numbers only, that's why this concept also called Real Number Systems.

↳ Any Number can be formed from digits only.

↳ digits [0, 1, 2, 3, 4, 5, 6, 7, 8, 9] 10 are taken from "Hindu Arabic System".

↳ The important Number types in Number Systems are :-

Whole Numbers:- Numbers from 0, 1, 2, - - - ∞

Natural Nos/ Counting No.s:- 1, 2, 3, 4, - - - ∞

Odd No.s:- Which are not divisible by 2.

Even No.s:- No.s which are exactly div by '2'.

Prime No.s:- Numbers which are divisibley 1 & 9tself (or) Numbers which are having only 2 factors.

Composite No.s:- Other than Prime No.s & Natural No.s excluding '1' are Composite (or) Having More than 2 factors.

Note:- '1' is Neither Prime Nor Composite

↳ Cross checking of Prime (or) Not:-

(i) Prime Number is always in the form of $6x \pm 1$. where x is natural Number. but not vice versa.

(ii) To find out prime (or) not, first find the nearby perfect square square root.

↳ And take the less than prime numbers to 9t

↳ Cross check whether these numbers devides the given number exactly (or) not

↳ If yes then composite otherwise Prime.

Ex:- 101 is prime (or) not ?

(i) 9ts in the form of $6x \pm 1$ $[6 \times 17 - 1]$.

(ii) Nearby perfect square is 100

less than prime No.s to $\sqrt{100} = 10$ are 2, 3, 5, 7.

No one from 2, 3, 5 & 7 devides 101 exactly then '101' is a prime No.s.

1 2 3 4 5 6 7 • 8 9 1 2 3

10^4 10^3 10^2 10^1 10^0 10^{-1} 10^{-2} ...

↳ Place position:- In which place digit is present [either before decimal (or) after decimal]

↳ before decimal place positions are $10^0, 10^1, 10^2$ ....

after decimal place positions are 10^{-1} [$\frac{1}{10^1}$], 10^{-2} ....

↳ Place value:- value at a particular place position.

Place value = Place position x digit @ that place.

Ex:- '3' [left to decimal] Place value:-

Place value of '3' is = Place position x digit

$= 10^4 \times 3 = 30,000$.

↳ Face Value:- Face value of a digit nothing but itself value only.

→ Divisibility Rules:-

1) '1':- It must be a number to divisible by '1'.

2) '2':- $2 = 2^{①}$ - Last '1'

Last digit (or) Unit digit must be divisible by '2' (or) Unit digit is either 0/2/4/6/8.

3) '3':- Sum of digits must be divisible by '3'.

4) '4':- $4 = 2^{②}$ → Last 2

Last '2' digits must be divisible by '4'.

5) '5':- Last digit is either '0' (or) '5'.